



Internship Project — Week 1

Project Title: House Price Prediction

Write a short paragraph (5–8 lines) inside a Markdown cell in your notebook answering these four points:

1. Which features influence house price the most?

Based on the correlation heatmap, the features that most strongly influence house price are **area**, **bathrooms**, and **stories**. These variables show the highest positive correlation with the price, meaning as these features increase, the property value consistently increases.

2. How accurate was your model (in plain terms)?

The Linear Regression model achieved an R^2 score of **0.65**. This means the model explains 65% of the variation in house prices. While this shows the model has good predictive power, it also suggests that there are other factors not present in this dataset that influence the remaining 35% of the price variation.

3. What surprised you in the data?

What surprised me most was that the total **area** of the property had a much stronger influence on the final price than the number of **bedrooms**. I initially assumed that the number of bedrooms would be the primary driver, but the data showed that square footage (area) and the number of bathrooms are much more significant predictors of value.

4. One recommendation for a real estate business based on your findings

Based on this analysis, real estate firms should prioritize listings with higher square footage and more bathrooms, as these are the strongest drivers of price. These properties should be the focus of premium marketing campaigns as they yield higher market valuations.